Print sheet

Generated 2026-11-11 00:10. Bed 256 x 256 x 260 mm (Bambu X2D). Masses and times are rough estimates: walls and infill per material from the settings below, volume at a typical print rate plus 2.14286 s per 0.2 mm layer, the layer time counted once per plate.

Parts

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
disp_aim_housing	PLA	1	176.5 x 59.0 x 183.0	212.8	82.4	1.77	761	yes, tree supports from the plate (761 mm2 over 45 deg)
disp_rail_mid	PLA	1	92.5 x 68.0 x 170.0	201.1	80.4	1.71	2333	yes, tree supports from the plate (2333 mm2 over 45 deg)
disp_big_gear	PLA	1	105.0 x 105.0 x 27.3	184.2	48.4	0.80	644	yes, tree supports from the plate (644 mm2 over 45 deg)
disp_rail_rear	PLA	1	180.9 x 68.0 x 76.1	150.9	63.0	1.17	4532	yes, tree supports from the plate (4532 mm2 over 45 deg)
disp_rail_front	PLA	1	131.5 x 128.0 x 48.5	146.5	64.4	1.11	7890	yes, tree supports from the plate (7890 mm2 over 45 deg)
disp_ebox	PLA	1	124.0 x 124.0 x 50.0	90.2	95.1	1.57	575	yes, tree supports from the plate (575 mm2 over 45 deg)
disp_motor_plate	PLA	1	104.0 x 114.0 x 10.0	77.2	32.1	0.51	0	no
disp_motor_leg	PLA	1	72.0 x 76.0 x 137.0	69.9	28.4	0.83	9	no
disp_axle	PLA	1	203.3 x 26.0 x 25.2	63.0	20.2	0.38	2321	yes, tree supports from the plate (2321 mm2 over 45 deg)
disp_drive_gear	PLA	1	70.0 x 69.7 x 28.8	43.5	21.0	0.40	3859	yes, tree supports from the plate (3859 mm2 over 45 deg)
disp_sleeve	PLA	1	64.0 x 64.0 x 85.0	39.4	19.6	0.55	2907	yes, tree supports from the plate (2907 mm2 over 45 deg)
disp_aim_crank	PLA	1	62.3 x 59.6 x 38.9	33.6	14.5	0.33	1756	yes, tree supports from the plate (1756 mm2 over 45 deg)
disp_carriage_rear	PLA	1	144.0 x 59.4 x 43.0	29.7	31.8	0.60	4474	yes, tree supports from the plate (4474 mm2 over 45 deg)
disp_yoke	PLA	1	43.3 x 126.6 x 34.0	21.0	15.2	0.33	1782	yes, tree supports from the plate (1782 mm2 over 45 deg)
disp_carriage_front	PLA	1	144.0 x 32.4 x 43.0	19.2	19.9	0.43	2197	yes, tree supports from the plate (2197 mm2 over 45 deg)
disp_aim_worm	PLA	1	28.0 x 28.0 x 54.5	17.1	9.3	0.30	2779	yes, tree supports from the plate (2779 mm2 over 45 deg)

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
disp_aim_lug	PLA	1	30.0 x 58.0 x 27.8	13.6	7.0	0.19	995	yes, tree supports from the plate (995 mm2 over 45 deg)
disp_aim_conrod	PLA	1	173.0 x 17.0 x 6.0	11.3	7.2	0.12	0	no
disp_gear_pin	PLA	1	53.0 x 16.0 x 12.5	4.3	2.4	0.07	68	no
disp_aim_cap	PLA	1	45.5 x 14.0 x 19.4	4.2	3.9	0.12	0	no
disp_anchor_pin_l	PLA	1	8.0 x 45.0 x 8.0	2.2	1.4	0.04	108	yes, tree supports from the plate (108 mm2 over 45 deg)
disp_anchor_pin_r	PLA	1	8.0 x 45.0 x 8.0	2.2	1.4	0.04	108	yes, tree supports from the plate (108 mm2 over 45 deg)
disp_aim_pin	PLA	1	15.0 x 13.0 x 30.3	2.0	1.3	0.11	0	no

disp_aim_housing: As modelled, bottom face on the bed. Aim housing: stands on the board, carries the sleeve end
disp_rail_mid: Rotated: the part's -x axis points up. Rail mid piece (cheeks on the sleeve, big gear pin bosses); glue the front joint (no splice straps)
disp_big_gear: Rotated: the part's -z axis points up. Big gear; one M3 screw through the cheek fixes it with its first tooth on the rack
disp_rail_rear: As modelled, bottom face on the bed. Rail rear piece with the tether pulley, its bearing, washers and axle fused in
disp_rail_front: As modelled, bottom face on the bed. Rail front piece (spring anchor posts, muzzle tie)
disp_ebox: As modelled, bottom face on the bed. Electronics box (D50 bay, thin walls, no lid): ESP32-S3 DevKitC-1 cradle with USB opening, BTS7960 on 4 standoffs (M3 inserts or self tap), DRV8833 and MP1584 pockets with cable-tie slots
disp_motor_plate: Rotated: the part's +y axis points up. Motor adapter plate: the real wiper motor bolts on from behind with 3 x M6 (bosses on the rear face, slots +-4 mm: the D50 pattern is assumed, not measured); its shaft goes through the 30 mm hole into the axle hub
disp_motor_leg: As modelled, bottom face on the bed. Motor plate leg: the plate drops into its 10 mm slot (glue); 3 screws into the board
disp_axle: Rotated: the part's +z axis points up. Main axle: D50 coupler hub (the real wiper shaft plugs in) + main shaft, round, d12 spigot into the sleeve. Prints lying on its flat, support under the spigot
disp_drive_gear: As modelled, bottom face on the bed. Drive gear 26T with the hub fused in (keyway onto the sleeve rib)
disp_sleeve: Rotated: the part's -z axis points up. Launch sleeve: the rail pivot on the main axle
disp_aim_crank: As modelled, bottom face on the bed. Worm wheel and crank with the two 608 bearings fused in (runs on the printed pin)
disp_carriage_rear: Rotated: the part's +y axis points up. Carriage (rack, cradle, pusher, lip roller fused) in two halves: glue at the cut face
disp_yoke: Rotated: the part's -z axis points up. Carriage yoke with both spring pegs fused on (the real spring hooks slip onto the peg ends)
disp_carriage_front: Rotated: the part's +y axis points up. Carriage (rack, cradle, pusher, lip roller fused) in two halves: glue at the cut face
disp_aim_worm: As modelled, bottom face on the bed. Worm: self-locking, it holds the crank and so the rail angle
disp_aim_lug: Rotated: the part's +y axis points up. Conrod lug on the left rail wall (2 x M3 or glue)
disp_aim_conrod: Rotated: the part's +y axis points up. Conrod
disp_gear_pin: Rotated: the part's -x axis points up. Big gear pin (no clip: a drop of glue)
disp_aim_cap: Rotated: the part's -z axis points up. Worm saddle cap
disp_anchor_pin_l: Rotated: the part's +y axis points up. Spring anchor pin (printed; or 8 mm rod)
disp_anchor_pin_r: Rotated: the part's +y axis points up. Spring anchor pin (printed; or 8 mm rod)
disp_aim_pin: Rotated: the part's -z axis points up. Crank axle pin (printed in place of the M8 bolt)

Plates

plate	material	objects	mass g	time h
plate_1_PLA.3mf	PLA	6	177	2.9
plate_2_PLA.3mf	PLA	8	278	4.7
plate_3_PLA.3mf	PLA	9	216	3.8

Totals on the plates

material	objects	plates	mass g	filament m	time h
PLA	23	3	670	224.7	11.3

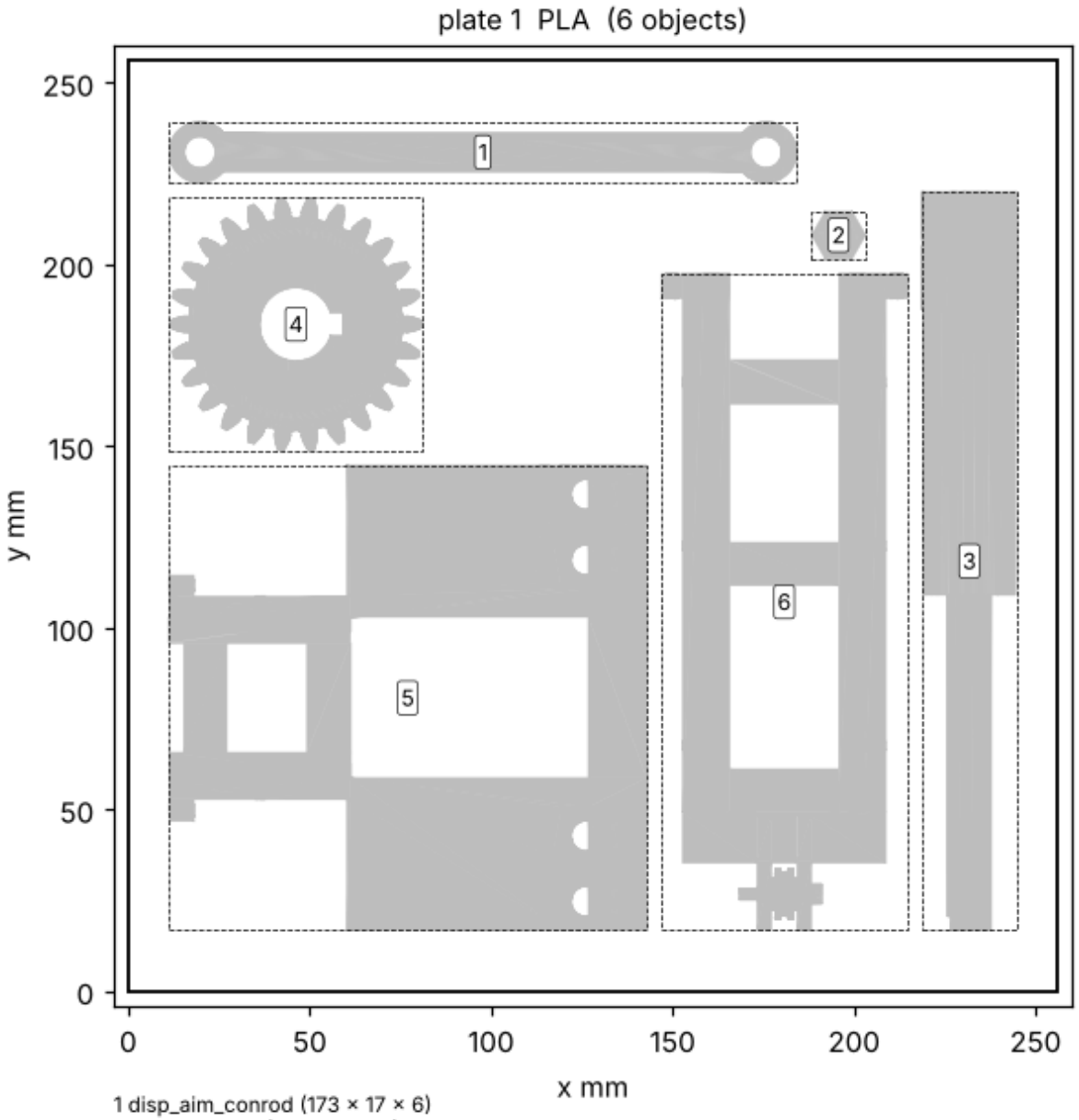
Recommended settings

PLA

- 0.4 mm nozzle, 0.2 mm layers
- 4 walls, 5 top and 5 bottom layers
- 40 % gyroid infill
- Nozzle 220 C, bed 55 to 60 C, textured PEI plate
- Door open or top vented

plate_1_PLA.3mf

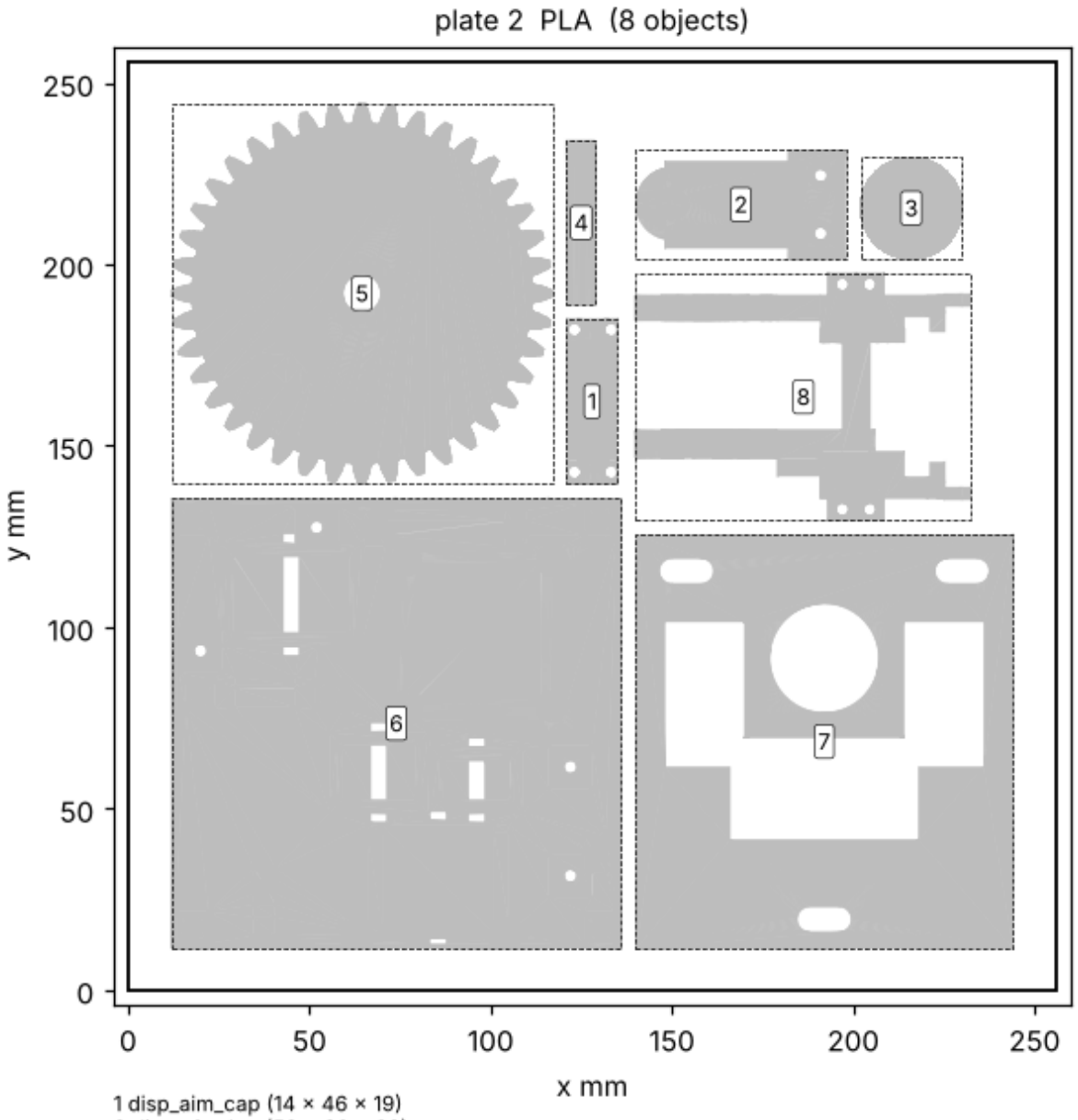
PLA, 6 objects, about 177 g and 2.9 h.



1 disp_aim_conrod (173 × 17 × 6)
2 disp_aim_pin (15 × 13 × 30)
3 disp_axle (26 × 203 × 25)
4 disp_drive_gear (70 × 70 × 29)
5 disp_rail_front (132 × 128 × 48)
6 disp_rail_rear (68 × 181 × 76)

plate_2_PLA.3mf

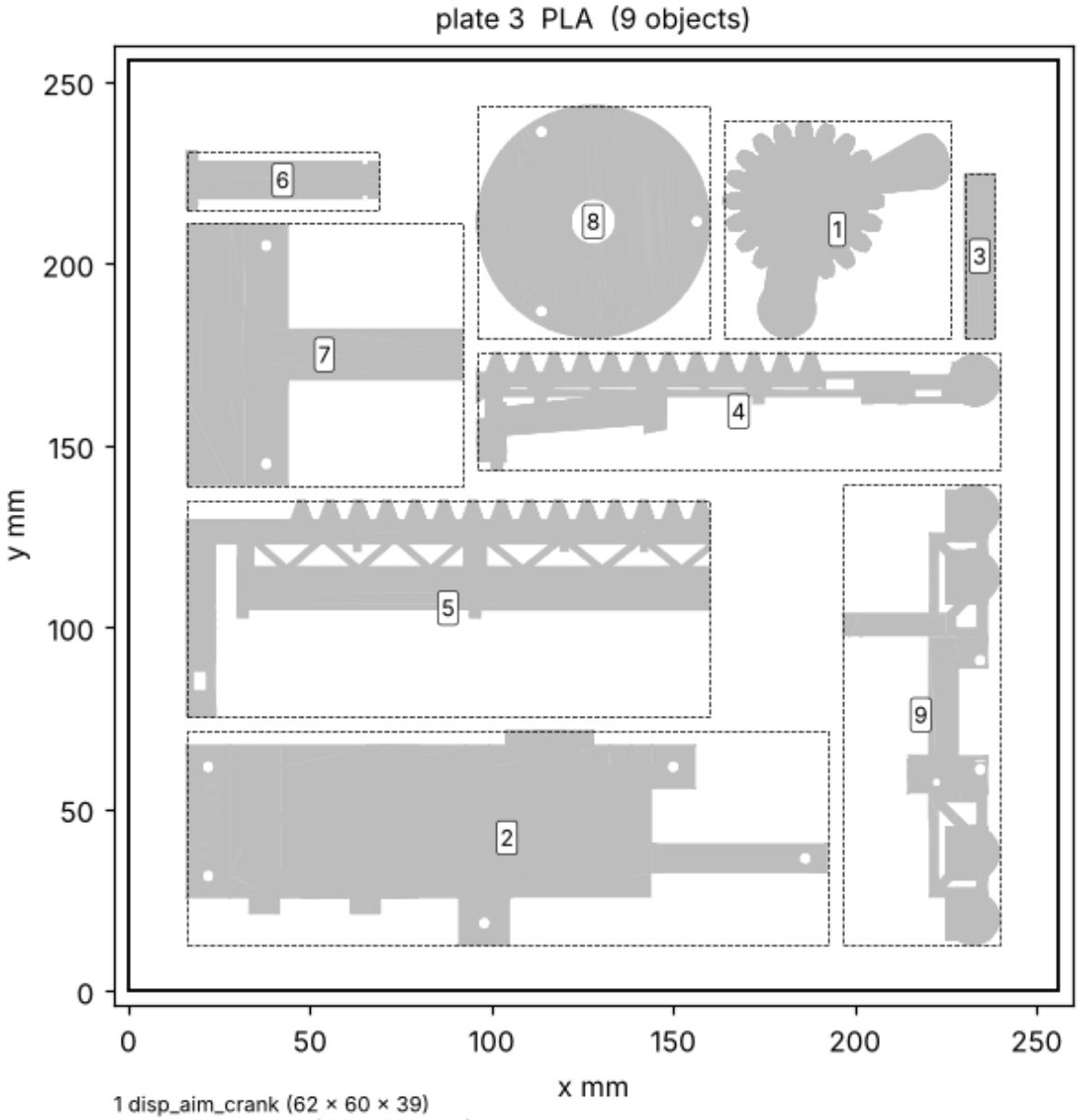
PLA, 8 objects, about 278 g and 4.7 h.



1 disp_aim_cap (14 × 46 × 19)
2 disp_aim_lug (58 × 30 × 28)
3 disp_aim_worm (28 × 28 × 54)
4 disp_anchor_pin_r (8 × 45 × 8)
5 disp_big_gear (105 × 105 × 27)
6 disp_ebox (124 × 124 × 50)
7 disp_motor_plate (104 × 114 × 10)
8 disp_rail_mid (92 × 68 × 170)

plate_3_PLA.3mf

PLA, 9 objects, about 216 g and 3.8 h.



1 disp_aim_crank (62 × 60 × 39)
2 disp_aim_housing (176 × 59 × 183)
3 disp_anchor_pin_l (8 × 45 × 8)
4 disp_carriage_front (144 × 32 × 43)
5 disp_carriage_rear (144 × 59 × 43)
6 disp_gear_pin (53 × 16 × 12)
7 disp_motor_leg (76 × 72 × 137)
8 disp_sleeve (64 × 64 × 85)
9 disp_yoke (43 × 127 × 34)